# Why Leaves Change Color

Name: _____ Date:

**INSTRUCTIONS:** Read the passage carefully. Then circle the best answer for questions 1 to 4 and write full-sentence answers for questions 5 to 7.

In summer, leaves are green because they are full of a chemical called chlorophyll. Chlorophyll catches sunlight and uses it to turn water and air into sugar, which feeds the tree. This process is called photosynthesis. Chlorophyll is so plentiful in summer that it hides every other color in the leaf.

As autumn arrives, the days get shorter and the nights get cooler. The tree senses the change and begins to shut down its leaves for winter. It stops making chlorophyll, and the green fades away. Now the colors that were hiding all along can be seen. Yellow and orange come from chemicals called carotenoids, which were in the leaf the whole summer.

Red and purple are different. They are made fresh in autumn from sugar that gets trapped in the leaf. Bright sunny days and cool, but not freezing, nights make the most sugar, so those years give the reddest leaves. A warm, cloudy autumn gives duller colors.

Finally, a thin layer of cells grows across the base of each leaf stem, like a tiny wall. The leaf can no longer get water, it dries out, and a gust of wind carries it to the ground. The tree rests through the winter and grows new green leaves in spring.

## CIRCLE THE BEST ANSWER

1. What makes leaves green in summer?

   a) Sunlight b) Chlorophyll c) Sugar d) Cold nights

2. Where do the yellow and orange colors come from?

   a) They are made new in autumn. b) Chemicals that were in the leaf all summer.

   c) Frost on the leaf. d) The soil around the tree.

3. Which weather gives the reddest leaves?

   a) Warm and cloudy b) Rainy and windy c) Sunny days and cool nights d) Freezing nights

4. In the first paragraph, "plentiful" means

   a) rare b) present in large amounts c) green d) dangerous


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## WRITE YOUR ANSWER IN FULL SENTENCES

5. What is photosynthesis?

6. Why does a leaf finally fall off the tree?

7. Write the main idea of the whole passage in one sentence.

## DRAW AND LABEL THE SAME LEAF IN TWO SEASONS

Summer leaf (label the chemical that makes it green)

Autumn leaf (label where each color comes from)



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## ANSWER KEY

### CIRCLE THE BEST ANSWER

1. b) Chlorophyll

2. b) Chemicals that were in the leaf all summer.

3. c) Sunny days and cool nights

4. b) present in large amounts

### WRITE YOUR ANSWER IN FULL SENTENCES

5. The process in which chlorophyll uses sunlight to turn water and air into sugar that feeds the tree.

6. A thin layer of cells grows across the base of the stem and cuts off the water, so the leaf dries out and the wind carries it away.

7. Leaves change color in autumn because the green chlorophyll fades, showing the yellows and oranges that were always there, while reds are made from trapped sugar.

### DRAW AND LABEL THE SAME LEAF IN TWO SEASONS

Summer: chlorophyll.
Autumn: yellow/orange from carotenoids, red/purple from trapped sugar.


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